

Drift of the Uncharted: A Speculative Fiction About Rising Sea-Level Landscapes through Robotic Performance

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Figure 1: Exhibition view of *Drift of the Uncharted*

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Abstract

Drift of the Uncharted presents as a robotic performance in combination with expanded cinema. It combines climate prediction and real-world landscape reconstruction, revealing a speculative world where contemporary urban landscapes experience future sea-level rise. A quadruped robot with an onboard projector serves as a rescue robot, simultaneously traversing both physical and virtual spaces, projecting the drowned cityscapes into the exhibition space. This research examines how speculative narrative and robotic performance mediate climate change—a hyperobject that transcends conventional scales of space and time into an experiential form

while simultaneously expanding robotic art by positioning the robot as a narrative agent.

CCS Concepts

• **Applied computing** → **Media arts**.

Keywords

Robotic Art, Agent-based Art, Climate Change

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1 Introduction

According to the IPCC's Sixth Assessment Report (AR6), global average temperatures have risen by approximately 1.1°C since industrialization, leading to increased frequency and intensity of extreme weather events [Intergovernmental Panel on Climate Change 2023]. Among the most pressing risks caused by global warming is coastal flooding, which threatens to inundate densely populated urban regions, disrupt infrastructure, displace communities, and accelerate ecological degradation in critical coastal ecosystems [Hinkel et al. 2014]. Despite the growing clarity of scientific evidence, public awareness and actions regarding the climate crisis remain insufficient, reflecting broader societal challenges in addressing systemic environmental issues. As illustrated by Timothy Morton's concept of hyperobjects, climate change operates at scales and temporalities that transcend individual experience, making it difficult to directly perceive or represent [Morton 2013]. We often rely solely on localized climate phenomena, data, and models to construct an understanding of this vast climatic puzzle. This project aims to explore how artistic practice can mediate such elusive and perceptually challenging concepts of climate change into experiential encounters accessible to audiences.

Drift of the Uncharted (DOU) focuses on sea-level rise as a critical manifestation of climate change, constructing speculative narratives through climate-sea level predictions within the framework of computational culture (Fig1). The project draws its conceptual foundation from Climate Central's Coastal Risk Screening Tool (CRST), a web-based mapping tool that overlays projected sea-level rise scenarios with elevation data, based on IPCC AR6 models and coastal flood risk analysis. CRST integrates satellite-based elevation models with localized flood projections, identifying geographical areas highly vulnerable to submersion by mid-century if current emissions continue, providing fertile ground for speculative climate future [Climate-Central 2022]. We scan and reconstruct these low-lying areas based on CRST projections (Fig.3), integrating climate predictions, on-site 3D reconstructions, game engine technologies, and a quadruped robot to weave an experiential speculative fiction of climate change. To navigate the mixed-reality environment, we employ a quadruped robot fitted with a mobile projector and spatial tracker. Its physical movements control a virtual camera in real time, tightly linking bodily action with movement within the

virtual world. As it slowly moves through the darkened space, the robot acts as a roaming spotlight, unveiling fragmented glimpses of cities submerged by rising seas. The robot is framed as a rescue dog-like agent, virtually surveying post-disaster zones shaped by climate catastrophe. This choice draws on the long-standing cultural role of robots as human surrogates in high-risk or extreme environments, from disaster response units to exoplanetary exploration. This proxy structure repositions the audience—not as autonomous explorers, but as followers of a non-human agent navigating a cityscape shaped by uncontrollable forces. The resulting passivity reflects the emotional conditions of climate change as a hyperobject: uncertainty, disorientation, and helplessness.

In this paper, we first introduce the creation process of DOU, outlining its development from concept to the construction of a mixed reality based on climate predictions, comprising 3D reconstructions of real-world sites and CGI sea-level imagery—and the implementation of synchronised robot behavior control and mobile projection mapping. We situate the work within the context of media arts, examining how climate forecasting, spatial data, and robotic performance converge to form a techno-cultural interaction that culminates in a mixed reality experience. Furthermore, we analyze the cultural and behavioral transformation of robotic agency, exploring how the autonomous performance and behavioral aesthetics of the robot in DOU reinforce its speculative narrative and contribute new dimensions to contemporary discourse on agent-based aesthetics.

2 Related Works

A major barrier to mobilizing public action on climate change is psychological distance, a notion in social psychology that parallels Timothy Morton's idea of the hyperobject—both describe phenomena whose vast spatio-temporal scales exceed direct human perception and thereby hinder action [Maiella et al. 2020; Morton 2013]. Research in virtual reality, indicates that presence can counteract psychological distance by confronting audiences with remote phenomena such as glacial melt [Thoma et al. 2023]. Artistic practices extend this approach by creating immersive environments in shared physical spaces to deepen embodied perception. Clea T. Waite's *Ice-Time* uses expanded cinema to reveal glacial transformations across micro to planetary scales, immersing viewers in otherwise inaccessible polar environments [Waite 2023]. Superflux's *Mitigation of Shock* presents a hyper-detailed 2050 London apartment, situating visitors in the textures, food options, routines, and constraints of climate-adapted life [Superflux 2017]. DOU advances this lineage through mixed reality, inviting embodied observation of speculative climate futures as drowned cityscapes.

Another DOU trajectory builds on the agent-based art by Simon Penny, conceptualizing the agent's behavior as an aesthetic domain [Penny 2017]. Gilberto Esparza's *Nomadic Plant* and Ana Laura Cantera's *Invisible Cartographies* send robots into natural environments for air quality monitoring or the remediation of polluted water sources, merging ecological engagement with robotic agency to evoke a cross-species environmental imaginary [Cantera 2017; Thompson and Mukhopadhyay 2014]. DOU situates robotic behavior within a mixed reality context, inviting viewers to perceive it as a human proxy exploring submerged cityscapes.



Figure 2: Close-up view of DOU

3 Concept Framework in Drift of the Uncharted

Speculative narrative in DOU envisions possible future coastal landscapes threatened by rising sea levels, using predictions from Climate Central's CRST based on IPCC data. We scanned vulnerable areas in three cities using 3D Gaussian Splatting (3DGS) to reconstruct them and simulate flooding in Unity. A quadruped robot equipped with a projector navigates physical space while traversing and projecting the virtual future environment (Fig.2).

3.1 Mediating the Hyperobject: Sea-Level Rise in a Mixed-Reality Experience

Timothy Morton characterises climate change as a hyperobject—its vast scales exceeding the limits of embodied and individual perception [Morton 2013]. DOU engages with this hyperobject through the concrete phenomenon of sea-level rise, which threatens coastal cities by reshaping landscapes, compromising infrastructure, and degrading ecosystems like wetlands. The speculative narrative begins with CRST, which overlays IPCC 2021 climate projections onto global elevation data to visualize regions likely to be below sea level in the near future (2050–2100) [Climate-Central 2022], forming the basis for our inquiry. As Paul N. Edwards notes, climate forecasts are constructed from data, sensors, models, and social

infrastructure—not reality itself [Edwards 2010]. Likewise, CRST cannot account for local factors like flood defences, but it opens space for artistic speculation: What could a world transformed by rising seas look like? DOU operates within this uncertainty, blending scientific prediction, real-world reconstruction, and artistic agency to create a mixed-reality experience:

Based on CRST projections, we selected Shanghai, Shenzhen, and Guangzhou—China's most dynamic and densely populated coastal cities and among the most vulnerable to future sea-level rise (Fig.3). The Bund's iconic skyline and Lujiazui's financial district, once bathed in golden sunlight through plane-tree shadows, may face the threat of inundation in the near future. To evoke nostalgia in the audience, we conducted field visits to sites steeped in local memory—cultural centres and historic residences, meandering old alleys, and tranquil wetland parks, among others. Through field visits, on-site photography, and 3DGS, we reconstructed these scenes and overlaid CGI seawater in Unity Engine, transforming familiar urban scenes into post-flood landscapes.

DOU stages a mixed-reality experience driven by the robot's real-time behavior. As the robot traverses the gallery, it simultaneously explores flooded landscapes in Unity. Live projection fuses both spaces, situating the work beyond the single screen into an

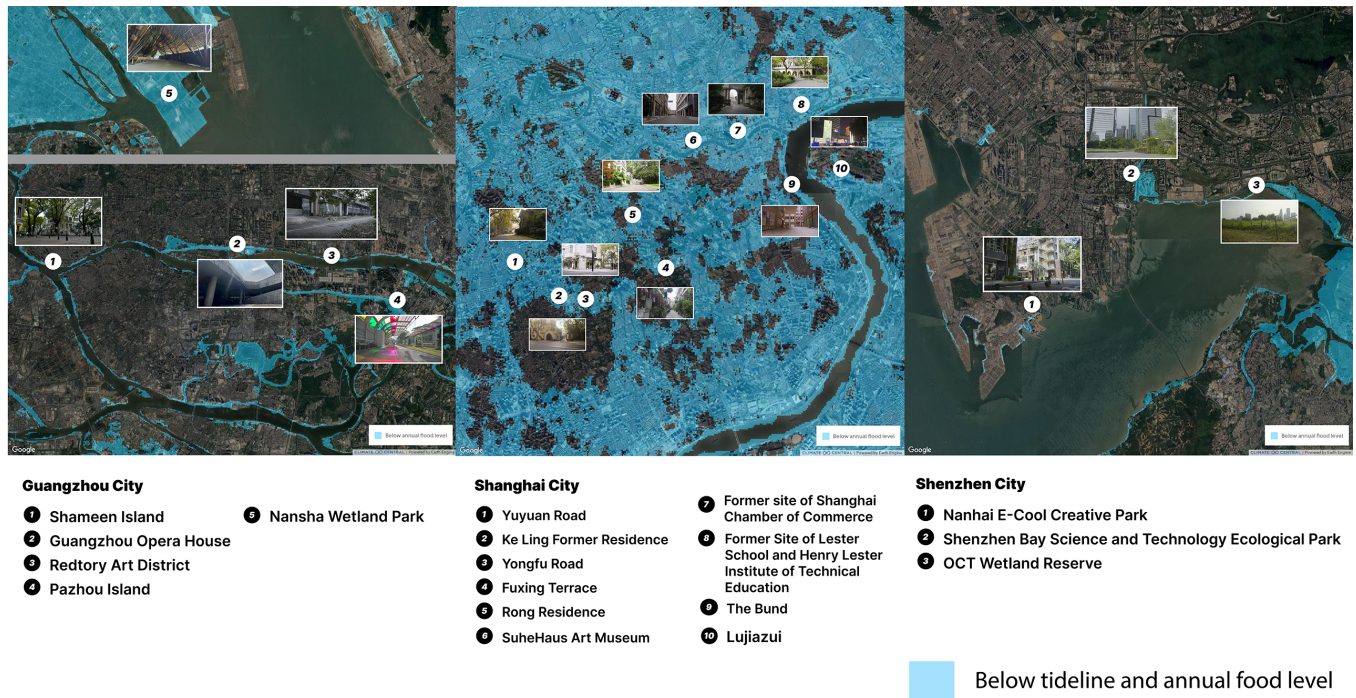


Figure 3: Scenes Selection

expanded cinema to activate the site [Youngblood and Fuller 2020], transforming the space into a cinematic speculative climate future. DOU extends this lineage by externalizing the camera as a robotic performer, where framing and montage are produced through behavior— speed, pauses, and head turns. It situates audiences within this layered reality: they perceive the tangible presence of the robot even as the same agent continually recasts the surrounding architecture into a dynamic, submerged world. DOU addresses Morton’s claim that hyperobjects exceed individual perception by treating artistic practice as a sensory amplifier. Meditating the hyperobject into an embodied experience, invites audiences to attune to future coastal risk.

3.2 Quadruped Robots: exploring climate future through a rescue agent Perspective

In DOU, a quadruped robot is framed as a human surrogate surveying areas submerged by sea-level rise. This choice follows a long-standing culture of robots in hazardous environments—from planetary exploration to deep-sea missions [Brantner and Khatib 2021; Chiappe and Vervaeke 2021]. A quadruped form factor was selected for its mobility on complex, partially inundated terrain. Boston Dynamics’ Spot, operating in the complex terrain of the Fukushima disaster zone, and Unitree quadrupeds, advertised for rough and semi-aquatic tasks, both indicate the suitability of this platform for autonomous surveying of submerged environments [Dynamics 2022; Unitree 2022]. Building on Simon Penny’s view that agent-based art depends on audiences projecting living metaphors onto machines [Penny 2017], DOU harnesses the cultural association of robots as human surrogates to invite viewers to read the quadruped

as a rescue agent and inhabit its perspective on a future shaped by sea-level rise.

The interplay between mixed reality and robotic autonomy in DOU resonates with Louis-Philippe Demers’ *Machine Performer* concept, where perceived intentionality arises not only from intrinsic autonomy but also from narrative staging that invites audiences to co-create the machine’s presence by attributing meaning to its behaviors [Demers 2016]. By projecting a submerged world from the robot’s point of view, mixed-reality creates a performative stage that transforms the robot into a rescue agent, its mundane movements—pauses, head tilts—into expressive gestures of exploration. The robot’s behavior is tightly coupled with virtual camera control, synchronising bodily movement with shifts in projected perspective. This embodied correlation seamlessly enhances the mixed-reality merging of the physical and virtual space, forges a synthesis between robotic behavior and cinematic projection, and thus elevates the robot’s presence and narrative significance.

The experience of DOU is deliberately shaped as a guided journey rather than a freely navigable one, a core narrative strategy. By intentionally withholding direct control, we invite the audience to follow a non-human agent passively into a future they cannot command, much like observing Mars through the lens of a rover. This mode of passive observation is designed to mirror the uncertainty, helplessness, and latent anxiety that define our relationship with climate change.

This guided journey is realized through the robot, which acts as a human proxy. The robot’s projection functions like a dynamic spotlight, allowing audiences to catch glimpses of the virtual world only by following its movements. Audiences access this speculative

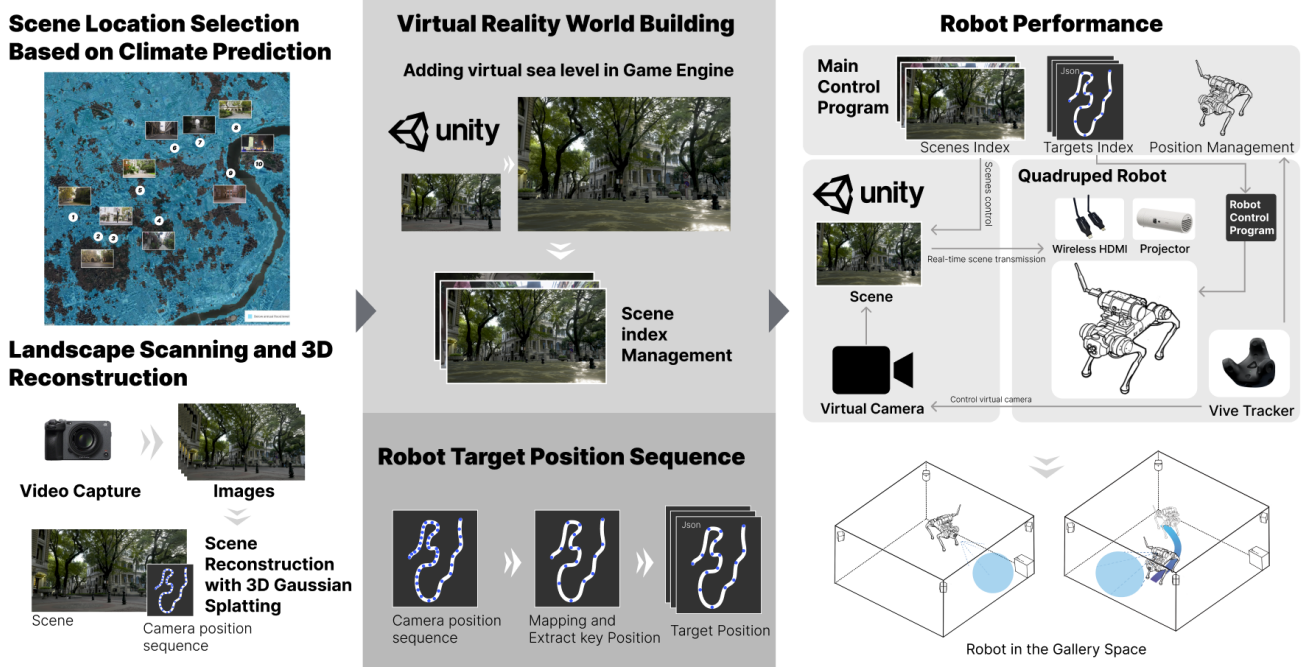


Figure 4: Main Pipeline

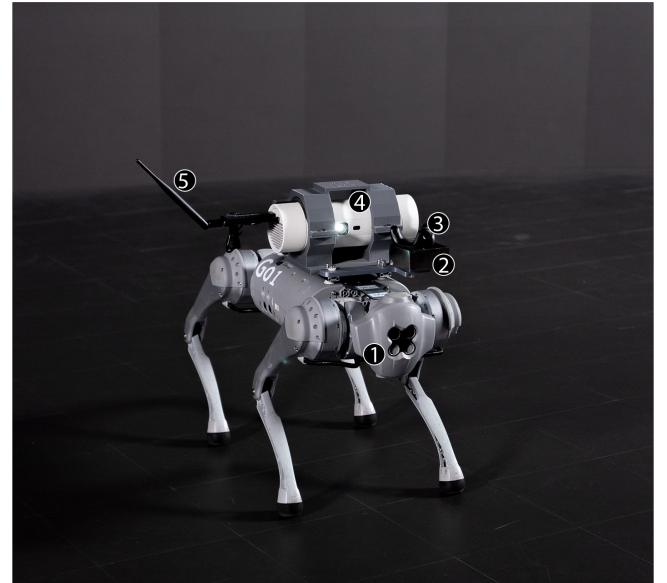
future by following the robot's perspective, piecing together a mental map from the projected fragments. The robot's performance is deliberately paced: it moves slowly, pauses at key points for random durations, and occasionally triggers head-turning gestures to simulate exploration. This slow, deliberate choreography, coupled with the sound of waves, creates a tranquil yet mysterious atmosphere for contemplating the speculative climate future—drowned cities.

4 Technical Implementation

This section delineates the technical architecture of DOU (Fig. 4), including the use of climate projections in the selection of real-world landscapes, reconstruction via 3DGS, and the synthesis of virtual world and projected seawater levels within Unity. Additionally, it explicates the quadruped robot's control system, which coordinates the virtual camera and facilitates the projection of the virtual world onto the physical site through its onboard projector.

4.1 Scene Location Selection Based on Climate Forecasting and Data Collection

Based on CRST 2050 projections, which are based on IPCC 2021 data and assume current emission policies leading to a projected 3.6°C temperature increase by 2100, we identified at-risk zones in Shanghai, Shenzhen, and Guangzhou. We then selected the specific, culturally significant sites detailed in the previous section for reconstruction (Fig.3). To reconstruct these locations, we used a 3DGS workflow with on-site video, enabling highly realistic 3D scene reconstruction from multi-view images, which offered greater fidelity and practicality than Lidar [Kerbl et al. 2023]. Crucially, since the final artwork presents a robot's-eye view, our data collection process had to mimic the robot's height and movement patterns



- ① Quadruped Robot
- ② Shock-resistant and anti-interference signal box
- ③ Vive Tracker
- ④ Projector with Built-in Battery
- ⑤ Wireless HDMI Receiver

Figure 5: Robot's Structure

to ensure a coherent reconstruction. We repurposed this technical constraint as an aesthetic strategy, effectively embodying the



Figure 6: Scenes Reconstruction and Virtual Seawater Integration

machine’s perspective during data acquisition using a Sony FX3 camera on an electric gimbal.

4.2 Scene Reconstruction and Virtual Seawater Integration

We processed our video footage using 3DGS to reconstruct more than 40 scenes, which generated Gaussian Splat scenes and corresponding camera trajectories. These assets were then imported into Unity and further refined. We added virtual seawater—matching its color to our field photography—to simulate the effects of sea-level rise (Fig.6).

To synchronize the robot with the virtual camera, we extracted a simplified sequence of key points from the reconstructed trajectory, containing precise position and orientation data. These key points guided the robot and ensured close alignment between its movements and the virtual camera.

4.3 Robot Performance Setup

Hardware Configuration, Tracking and Projection: Tracking and Projection: The setup includes a Unitree Go1 robot fitted with a Vive Tracker and an onboard projector, a control computer, and four Vive Base Stations (Fig.4-5). The Vive system relays the robot’s real-time position to the motion-control program and Unity through

the OpenVR SDK, enabling the robot’s movements to drive the virtual camera. The computer then streams the resulting imagery wirelessly to the robot’s projector, casting the virtual environment onto the exhibition space.

Behavior Control: The motion-control program manages a sequence of Unity scenes and corresponding robot target points. For each scene, it compares the Tracker’s real-time position with the target coordinates and transmits the data to the robot, which uses Unitree’s high-level system to adjust its orientation and movement. A behavioral layer further enhances performativity by enabling scripted travel speeds, randomized pauses at target points, and pre-defined gestures such as head movements.

5 Discussion and Conclusions

Penny argues that agent-based art shifts the aesthetic experience from viewing static objects to engaging with dynamic, relational behaviours [Penny 2017]. DOU draws the audience into a performative theatre shaped by the machine’s behavior, inviting them to assemble the narrative by following the robot’s exploration. This performative theatre is driven by the tight synchronization of the robot’s physical actions and its projected virtual world. Mixed-reality grounds the experience by anchoring the audience’s own body as the centre of perception. As they follow and observe, audiences experientially incorporate the machine’s perspective into

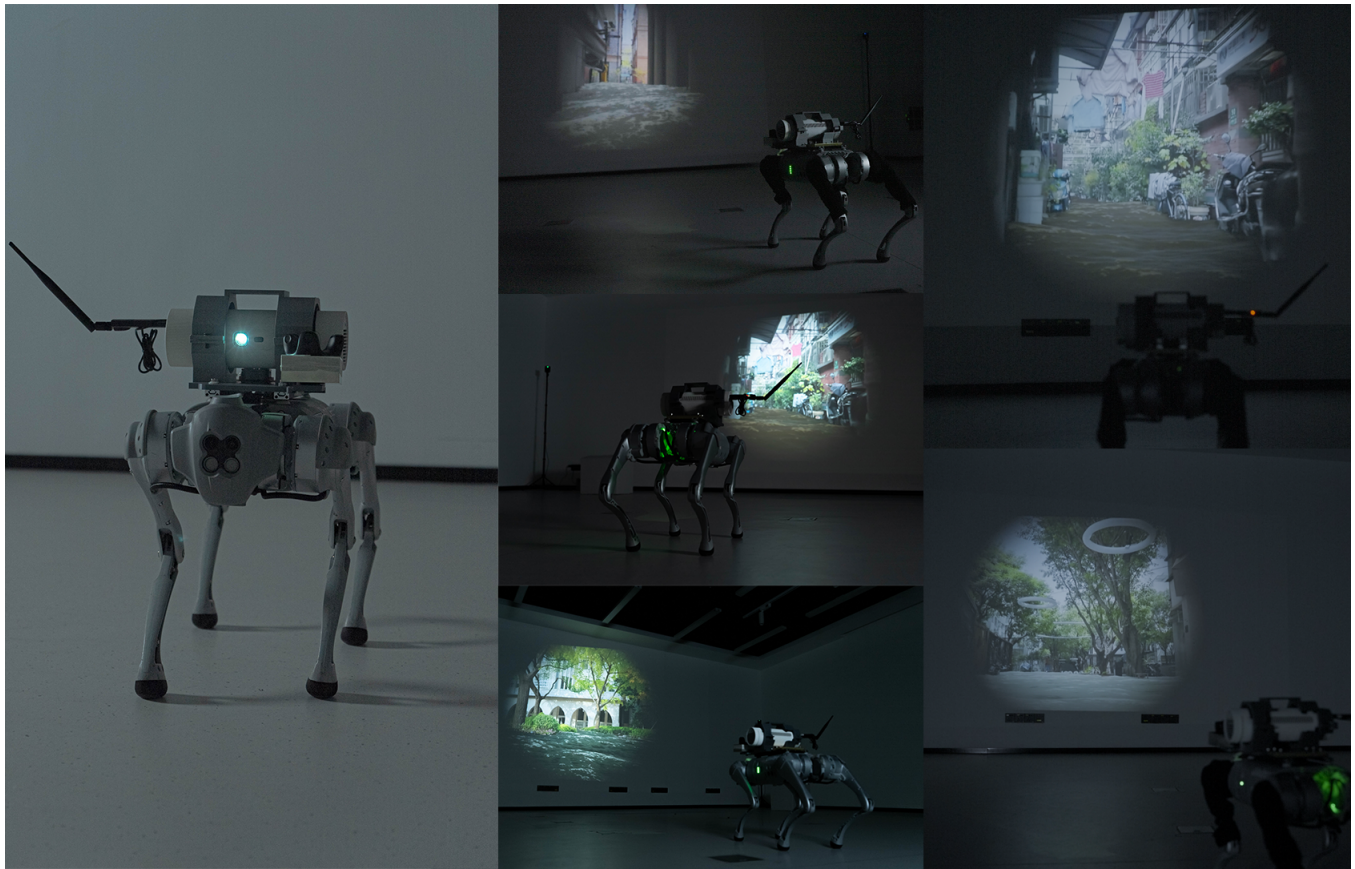


Figure 7: Quadruped robot in different scenarios

what Merleau-Ponty termed the 'body schema' [Merleau-Ponty and Landes 2013], with the robot functioning as an extended perceptual body. The result is a dual experience: viewers are aware of their physical co-presence with the robot yet are simultaneously guided by its non-human gaze into a virtual world (Fig.7).

Mixed-reality significantly extends DOU's performative potential as an agent-based artwork. Echoing Demers' concept of the 'machine performer,' DOU utilizes mixed reality to create a speculative climate future that functions as a narrative stage [Demers 2016]. The robot's simple mechanical movements—a pause or a turn—are transformed into intentional gestures like exploring or reflecting. This perceived intentionality is not inherent in the machine but is co-generated by the audience, who project meaning onto the robot's actions based on the synchronization of its behavior with its virtual point of view. The resulting dual presence is therefore reciprocal: the robot not only performs on the stage set by mixed reality, but its physical presence and behavior also serve to ground the virtual projection in a tangible, shared space. This ultimately suggests that for mixed reality to expand agent-based art, its potential lies not in simply providing context, but in achieving a deep, organic integration with the agent's core ontology of behavior and perception.

In conclusion, DOU synthesizes robotic performance and mixed reality to meditate the abstract hyperobject of climate change into a tangible, experiential narrative. The work establishes a perceptual framework through a non-human proxy—a robotic agent exploring a speculative flooded future. By following this agent, audiences experience the potential inundation of familiar landscapes, transforming abstract risk into a felt, personal threat. Thus, DOU not only offers a new model for an embodied experience of climate futures but also expands the boundaries of robotic art or agent-based art by repositioning the robot from a mere machine to a potent narrative and cultural mediator.

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